

Microscopic Transfer of Exact Meissner Weight

Infrared-stable Schrieffer–Wolff theory, a complete degree-eight no-go,
and an exact dark-branch finite-coupling completion

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Abstract

The exact pair-permutation QGN parent has a positive zero-frequency transverse Meissner weight although its fixed-number gap closes as L^{-2} . A screened multiorbital construction reproduces that parent at order $\lambda^6\Delta$ but leaves a Peierls-covariant quasi-local remainder at order $\lambda^8\Delta$. We resolve the resulting microscopic-transfer problem in three steps. First, target-metric deformations admit finite-volume static and imaginary-frequency response bounds independent of the closing parent gap, and their current defect is controlled in the H_{-1} topology at order $\lambda^{10}\Delta$. A bounded anisotropic metric nevertheless produces a finite Drude–Meissner mismatch, showing that row-relative form alone is not enough; hydrodynamic closure together with point-group scalar zero-momentum structure restores uniform H_{-1} tightness and yields $D_K = D_A = D_M$. Second, a complete weighted-degree-eight resolvent audit gives a closed source formula on the product-AGP manifold. A two-pair witness proves that the unmodified matched single-flavor and closed-shell Kramers families cannot preserve the complete composition manifold at that order. An explicit finite-range higher-body counterterm cancels the full unsafe source and restores the two-sided row ideal through degree eight, with response transfer conditional on the stated hydrodynamic audit. Third, the primitive bridge has an exact coherent dark channel $|\Omega_n^-\rangle \propto (\eta_1^+ - \eta_2^+)^n |0\rangle$. Strict local Schur-complement overcompensation makes it the unique fixed-number ground state of a positive-semidefinite finite-coupling family. In a point-group-complete $d \geq 2$ Peierls lift, the swap backbone yields the rigorous transverse floor $D_* = 2\lambda^6\Delta(j_1 + j_2)\rho(1 - \rho) > 0$. This is a finite-coupling zero-frequency microscopic Meissner lower bound for a selected coherent branch, not an all-composition transfer of the original matched family or an equality theorem for every gadget contribution.

Claim hierarchy.

1. The target-metric bounds, H_{-1} defect estimate, anisotropic counterexample, isotropic order-of-limits theorem, and local row-ideal criterion are exact transfer results and do not assume a uniform complete many-body gap.
2. The complete weighted-degree-eight class enumeration and source formula prove a no-go for preservation of the full product-AGP composition manifold in the unmodified matched single-flavor and Kramers families.
3. The explicit $\mathcal{C}_8$ counterterm gives a two-sided row-ideal completion through weighted degree eight; the Part I response conclusion additionally requires the stated hydrodynamic-closure and isotropy audit. Its synthesis from only the original restricted two-electron channels and an all-orders convergent completion remain open.
4. The coherent dark branch is an exact finite-coupling zero-field eigenbranch. It is not claimed to be the ground state of the original matched Hamiltonian.

5. The strictly overcompensated positive-semidefinite family has the dark AGP as its unique fixed-number ground state and has a rigorous positive transverse Kohn, Abelian, and Meissner *floor*. Equality of the total three kernels requires the Part I regularity conditions for the extra gadget rows.
6. All Meissner statements concern the zero-temperature matter current kernel. No dynamical Maxwell-field screening, finite-temperature transition, or real-axis conductivity decomposition is claimed.

1. Introduction and main results

For a selected finite-volume ground branch, let $\mathcal{C}_{L,T}(q)$ be the static curvature under a transverse Peierls field and let $\mathcal{K}_{L,T}(q, i\zeta)$ be the corresponding imaginary-frequency matter kernel. The three quantities relevant here are, schematically,

$$D_K = \lim_{L \rightarrow \infty} V_L^{-1} \mathcal{C}_L(q=0), \quad (1)$$

$$D_A = \lim_{\zeta \downarrow 0} \lim_{L \rightarrow \infty} V_L^{-1} \mathcal{K}_L(q=0, i\zeta), \quad (2)$$

$$D_M = \lim_{\substack{q \rightarrow 0 \\ q \cdot t = 0}} \lim_{L \rightarrow \infty} V_L^{-1} \mathcal{C}_{L,T}(q). \quad (3)$$

Kohn curvature, the Abelian zero-frequency uniform kernel, and the transverse Meissner kernel are distinct in general [1, 2]. The current-spectral H_{-1} criterion identifying the first two limits is developed in [5]. The exact pair-permutation parent is exceptional [6]: its transverse source lies in the target-space cokernel, so the finite-volume kernel is independent of both q and ζ on every co-closed field. At density ρ its electronic-link weight is

$$D_{\text{parent}} = 2j\rho(1 - \rho), \quad (4)$$

with one quarter of this value in the pair-phase convention $Q = 2A$.

The microscopic construction is perturbative. A local Schrieffer–Wolff block diagonalization [3, 7] gives

$$H_{\text{eff,L}}^{[7]}(A) = \lambda^6 \Delta H_{\text{par,L}}(A) + R_L^{(8)}(A), \quad \left\| \partial_A^r R_L^{(8)}(A) \right\|_\mu \leq C_{\mu,r} \Delta |\lambda|^8, \quad 0 \leq r \leq 2. \quad (5)$$

Here $\|\Phi\|_\mu := \sup_x \sum_{X \ni x} e^{\mu \text{diam } X} \|\Phi_X\|$ denotes the exponentially weighted quasi-local interaction norm for the interaction $\Phi = \{\Phi_X\}$; harmless fixed support-size factors may be absorbed into its constants. The parent gap closes quadratically, so the local norm estimate does not control the response by itself. Writing P for the exact parent zero projector and $Q = 1 - P$, a generic remainder creates two infrared-sensitive vectors,

$$QR^{(8)}P, \quad Q(\partial_A R^{(8)})P. \quad (6)$$

The first moves the zero-field branch; the second changes the current source. The paper resolves these two issues in three stages.

Part I: a sharp transfer framework. A two-sided parent-row factorization removes the state source exactly. The remaining current source can be controlled without a spectral gap by passing to target space. The resulting theorem gives a uniform perturbative floor, but an explicit anisotropic metric proves that bounded H_{-1} weight is not enough for exact temporal/transverse identification. Point-group isotropy supplies the missing power of momentum and yields $D_K = D_A = D_M$.

Part II: a complete degree-eight decision. The full canonical degree-eight coefficient of the unmodified screened family is enumerated and compressed to the complete product-AGP manifold. The resulting two-pair witness proves that all-composition transfer fails. This is stronger than the order-eight overlap remainder left unclassified in the microscopic construction [7], because every single-flavor class and the closed-shell Kramers correction are included. The same formula produces the exact finite-range counterterm needed to restore the Part I state-source hypothesis through degree eight. The current-sector closure and zero-momentum symbol remain a separate, finite audit.

Part III: a nonperturbative selected-branch route. The crossed bridge itself annihilates one coherent interblock AGP. Strict local overcompensation turns that branch into the exact ground state of a positive finite-coupling microscopic Hamiltonian. A direct-sum row argument then shows that no gadget relaxation can cancel the transverse swap-backbone source, which gives a positive microscopic Meissner floor before any small-gap perturbation theory is invoked.

The final claim boundary is therefore sharp. The unmodified matched family fails to preserve its complete composition manifold at weighted degree eight. A finite-order all-composition repair exists but is not yet synthesized from restricted two-electron channels or resummed to all orders. A modified positive family already has an exact finite-coupling dark ground branch and a rigorous zero-frequency transverse floor. The Part I theorem states the additional regularity needed to promote that floor to equality of the total Kohn, Abelian, and Meissner kernels.

Question	Result	Exact claim boundary
Can an $O(\lambda^8)$ local remainder be transferred through the closing parent gap?	Yes for bounded target-metric deformations; the static and imaginary-frequency kernels obey gap-independent bounds.	The zero-field state source must vanish, and the current source must factor through parent rows.
Is row-relative form alone enough for a Meissner identification?	No. The anisotropic metric gives a finite Drude–Meissner mismatch with bounded but nontight H_{-1} weight.	Exact identification needs soft-tail tightness, not only a uniform H_{-1} bound.
What closes the infrared limit?	Hydrodynamic closure plus a point-group scalar zero-momentum symbol gives $D_K = D_A = D_M$.	This is a sufficient theorem, not an automatic consequence of locality.
Does the unmodified matched family pass the local audit?	No. Its complete weighted-degree-eight coefficient leaks out of the product-AGP manifold at two pairs.	The no-go concerns analytic all-composition transfer near $\lambda = 0$.
What are the constructive routes?	$\mathcal{C}_8$ repairs the row ideal through degree eight; strict overcompensation selects an exact dark AGP and gives a finite-coupling floor.	The first route is finite order and higher body; the second is a selected-branch lower-bound theorem.

Part I. Infrared-stable transfer framework

2. Target-metric response stability

Let $D(t) : \mathcal{H} \rightarrow \mathcal{T}$ be a twice differentiable row map, where t is the amplitude of a chosen link field. Put

$$H_0(t) = \frac{1}{2} D(t)^\dagger D(t), \quad P = P_{\text{Ker } D(0)}, \quad S = \dot{D}(0)P. \quad (7)$$

For the exact transverse pair-permutation source,

$$D^\dagger S = 0. \quad (8)$$

Let $G(t)$ be a positive operator on the target space and define

$$H_G(t) = \frac{\varepsilon}{2} D(t)^\dagger G(t) D(t). \quad (9)$$

Since $DP = 0$, derivatives of $G(t)$ do not enter the first-order source on the zero manifold. If $\tilde{D} = G^{1/2} D$, then $\tilde{S} = G^{1/2} S$.

Theorem 2.1 (Weighted least-squares response bounds). *Assume*

$$0 < g_- \mathbf{1} \preceq G(0) \preceq g_+ \mathbf{1}. \quad (10)$$

For every vector in the zero manifold, the static curvature is the weighted least-squares residual

$$\langle \psi, \mathcal{C}_G \psi \rangle = \varepsilon \inf_{y \in \mathcal{H}} \langle S\psi - Dy, G(S\psi - Dy) \rangle. \quad (11)$$

If (8) holds, then for every $\zeta > 0$,

$$\boxed{\varepsilon g_- S^\dagger S \preceq \mathcal{C}_G \preceq \mathcal{K}_G(i\zeta) \preceq \varepsilon g_+ S^\dagger S.} \quad (12)$$

These inequalities are independent of the smallest positive singular value of D .

Proof. The ordinary positive-square response formula applied to $\tilde{D} = G^{1/2} D$ gives

$$\mathcal{C}_G = \varepsilon \tilde{S}^\dagger P_{\text{Ker } \tilde{D}^\dagger} \tilde{S}, \quad \mathcal{K}_G(i\zeta) = \varepsilon \tilde{S}^\dagger \frac{\zeta^2}{\tilde{\mathcal{L}}^2 + \zeta^2} \tilde{S}, \quad \tilde{\mathcal{L}} = \frac{\varepsilon}{2} \tilde{D} \tilde{D}^\dagger. \quad (13)$$

The first equality is equivalent to (11). Bounding the weighted norm above and below by $g_\pm$ and using $S \perp \text{Ran } D$ gives the first and last bounds. The middle inequality is the positive spectral defect between the finite-frequency multiplier and its zero-frequency projector. $\square$

For the microscopic scaling, set $\varepsilon = \lambda^6 \Delta$ and $\eta = \lambda^2$, and write

$$G_\eta = \mathbf{1} + 2\eta X, \quad X = X^\dagger, \quad 2\eta \|X\| < 1. \quad (14)$$

Then $g_\pm = 1 \pm 2\eta \|X\|$ are valid bounds. The entire transverse kernel is therefore changed by only $O(\varepsilon\eta) = O(\lambda^8 \Delta)$, uniformly in volume and frequency. This proves positivity at the precision needed by the microscopic expansion, but a sharper estimate identifies the exact infrared risk.

Theorem 2.2 (Gap-independent current H_{-1} estimate). *Under (8) and (14),*

$$J_G P = \varepsilon \eta D^\dagger X S. \quad (15)$$

Moreover, for every $\zeta > 0$,

$$\boxed{0 \preceq \mathcal{K}_G(i\zeta) - \mathcal{C}_G \preceq \frac{4\varepsilon\eta^2}{g_-} S^\dagger X^\dagger X S.} \quad (16)$$

If $D^\dagger X S = 0$, then

$$\mathcal{K}_G(i\zeta) = \mathcal{C}_G = \varepsilon S^\dagger G_\eta S \quad \text{for every } \zeta > 0. \quad (17)$$

Proof. Differentiating (9) and acting on P gives $J_G P = (\varepsilon/2) D^\dagger G_\eta S$. The term with the identity vanishes by (8), proving (15). The general spectral representation bounds the positive dynamical defect by twice the static paramagnetic form:

$$0 \preceq \mathcal{K}_G(i\zeta) - \mathcal{C}_G \preceq 2P J_G Q H_G^+ Q J_G P. \quad (18)$$

Because $H_G \succeq \varepsilon g_- H_0$ on Q ,

$$H_G^+ \preceq (\varepsilon g_-)^{-1} H_0^+. \quad (19)$$

Finally,

$$D H_0^+ D^\dagger = 2P_{\text{Ran } D}. \quad (20)$$

Substitution into (18) gives (16). If $D^\dagger X S = 0$, then $J_G P = 0$ and the source lies in $\text{Ker } \tilde{D}^\dagger$, proving (17). $\square$

Corollary 2.3 (Uniform perturbative transverse floor). *Let $D_{G,L}(\zeta, q_T)$ denote the volume-normalized kernel obtained from $\mathcal{K}_G(q_T, i\zeta)$ on a normalized allowed lattice-transverse source of momentum q_T , and let $D_{0,L}$ be the corresponding exact-parent density. Suppose $\sup_L D_{0,L} < \infty$. A bounded target-metric deformation satisfies*

$$\varepsilon(1 - 2\eta \|X\|) D_{0,L} \leq D_{G,L}(\zeta, q_T) \leq \varepsilon(1 + 2\eta \|X\|) D_{0,L} \quad (21)$$

for every finite L , every transverse field in the exact source-kernel class, and every $\zeta > 0$. Hence

$$D_G = \lambda^6 \Delta D_0 + O(\lambda^8 \Delta), \quad (22)$$

and the response remains positive for sufficiently small fixed λ on every branch with $D_0 > 0$.

The corollary gives a uniform positive transverse floor for a truncated response-safe low block. It does not, by itself, identify the exactly uniform Kohn/Drude limit with the transverse $q \rightarrow 0$ limit at fixed nonzero λ . The next example shows why an extra spatial condition is unavoidable.

3. A row-relative counterexample to exact order-of-limits transfer

Consider the gradient row on a two-dimensional torus. At momentum $\mathbf{q} = (q, 0)$ its symbol is

$$d(\mathbf{q}) = \begin{pmatrix} e^{iq} - 1 \\ 0 \end{pmatrix}, \quad t(\mathbf{q}) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad d(\mathbf{q})^\dagger t(\mathbf{q}) = 0. \quad (23)$$

Take the bounded anisotropic target metric

$$G_\eta = \begin{pmatrix} 1 & 2\eta \\ 2\eta & 1 \end{pmatrix}, \quad 0 < \eta < \frac{1}{2}. \quad (24)$$

This is an exact row-relative perturbation, preserves the zero state, and satisfies all volume-uniform norm bounds. Nevertheless,

$$T = t^\dagger G_\eta t = 1, \quad E(\mathbf{q}) = \frac{1}{2} d^\dagger G_\eta d = \frac{1}{2} |e^{iq} - 1|^2, \quad J(\mathbf{q}) = \frac{1}{2} d^\dagger G_\eta t = \eta(e^{-iq} - 1). \quad (25)$$

Therefore

$$\mathcal{C}_\eta(\mathbf{q}) = 1 - 4\eta^2, \quad \mathcal{K}_\eta(\mathbf{q}, i\zeta) = 1 - 4\eta^2 \frac{E(\mathbf{q})^2}{E(\mathbf{q})^2 + \zeta^2}. \quad (26)$$

At exactly $q = 0$ the current source vanishes and the uniform Drude kernel is one. Taking $q \rightarrow 0$ at fixed $\zeta > 0$ also gives one, whereas the static transverse limit is $1 - 4\eta^2$. The current H_{-1} weight is

$$\frac{|J(\mathbf{q})|^2}{E(\mathbf{q})} = 2\eta^2, \quad (27)$$

independent of q . Thus it is bounded but not tight. In microscopic units the mismatch is $O(\varepsilon\eta^2) = O(\lambda^{10}\Delta)$: two orders smaller than the static $O(\lambda^8\Delta)$ uncertainty, but nonzero at fixed coupling.

By contrast, $G_\eta = (1 + 2\eta)\mathbf{1}$ gives $J(\mathbf{q}) = 0$ for every transverse $t(\mathbf{q})$ and preserves all three limits exactly. The distinction is whether the zero-momentum metric preserves the spatial vector decomposition.

4. Point-group isotropy gives uniform transverse H_{-1} tightness

We now formulate the sufficient condition relevant to a symmetric d -dimensional microscopic completion. Let $d(\mathbf{q}) \in \mathbb{C}^d$ be the lattice-gradient symbol and let $t(\mathbf{q}) \in \mathbb{C}^d \otimes \mathbb{C}^r$ be a normalized current-source vector satisfying

$$(d(\mathbf{q})^\dagger \otimes \mathbf{1}_r)t(\mathbf{q}) = 0. \quad (28)$$

The internal space allows several blocks or several vector-row copies.

Definition 4.1 (Hydrodynamically closed geometric row metric). Fix a selected ground branch and a sufficiently small nonzero lattice momentum $\mathbf{q}$. In the translation- $\mathbf{q}$ fiber define the current-cyclic subspace

$$\mathcal{H}_{\text{cyc}}(\mathbf{q}) = \overline{\text{span}} \{p(H_\eta(\mathbf{q}))QJ_T(\mathbf{q})P\xi : p \in \mathbb{C}[z], \xi \in P\mathcal{H}\}. \quad (29)$$

Call an orthogonal decomposition $\mathcal{H}_{\text{cyc}}(\mathbf{q}) = \mathcal{H}_{\text{soft}}(\mathbf{q}) \oplus \mathcal{H}_{\text{reg}}(\mathbf{q})$ admissible if both summands are invariant under $H_\eta(\mathbf{q})$. Let $D_{\text{reg},L}(\mathbf{q}, i\zeta; t)$ denote the complete kernel-density contribution left after subtracting the soft-sector response below, including any residual stress term. We require this regular contribution to be uniformly H_{-1} -tight, to possess a thermodynamic limit jointly continuous at $(\mathbf{q}, \zeta) = (0, 0)$, and to obey

$$|D_{\text{reg}}(\mathbf{q}, i\zeta; t)| \leq C\varepsilon\eta \|t\|^2 \quad (30)$$

near the origin. A target-metric deformation is *hydrodynamically closed* if there is such an admissible decomposition, a momentum-intertwining unitary

$$U_{\mathbf{q}} : \mathcal{H}_{\text{soft}}(\mathbf{q}) \longrightarrow \mathbb{C}^r \quad (31)$$

and a positive matrix symbol $G_\eta(\mathbf{q})$ such that, writing $\Pi_{\text{soft}}(\mathbf{q})$ for the soft projection, the compressed Hamiltonian, current source, and associated stress are exactly

$$U_{\mathbf{q}}H_\eta(\mathbf{q})\Pi_{\text{soft}}(\mathbf{q})U_{\mathbf{q}}^\dagger = E_\eta(\mathbf{q}) = \frac{\varepsilon}{2}(d^\dagger \otimes \mathbf{1})G_\eta(\mathbf{q})(d \otimes \mathbf{1}), \quad (32)$$

$$U_{\mathbf{q}}\Pi_{\text{soft}}(\mathbf{q})QJ_T(\mathbf{q})P = J_\eta(\mathbf{q}; t) = \frac{\varepsilon}{2}(d^\dagger \otimes \mathbf{1})G_\eta(\mathbf{q})t(\mathbf{q}), \quad (33)$$

$$T_\eta(\mathbf{q}; t) = \varepsilon t(\mathbf{q})^\dagger G_\eta(\mathbf{q})t(\mathbf{q}). \quad (34)$$

Thus the full selected-branch kernel density is the sum of the exact soft metric response determined by (32)–(34) and $D_{\text{reg},L}$. This formulation makes “represented by a symbol” an explicit momentum-fiber intertwining statement rather than a heuristic identification.

Example 4.2 (Exact scalar-convolution closure). Let the parent rows be the lattice gradient tensored with an internal row space, and let the target metric be a translation-invariant absolutely summable convolution,

$$(G\psi)_{i,\alpha}(x) = \sum_{y,j,\beta} g_{i\alpha,j\beta}(x-y)\psi_{j,\beta}(y). \quad (35)$$

The Bloch transform is the intertwiner in (31) and diagonalizes the row metric with symbol $G(\mathbf{q}) = \sum_x e^{-i\mathbf{q}\cdot x} g(x)$. On the density-generated soft fiber $d(\mathbf{q}) \otimes \mathbb{C}^r$, the three compressed operators are exactly (32)–(34). Thus a scalar convolution of the pair-permutation rows is not merely an analogy: it satisfies the closure definition operator-theoretically. Here $\mathcal{H}_{\text{reg}}(\mathbf{q}) = \{0\}$ and $D_{\text{reg}} = 0$.

Remark 4.3 (Finite audit required for an operator-valued SW remainder). For a completed microscopic $d \geq 2$ lift, the closure hypothesis is discharged by a finite sequence of checks: construct the translation- $\mathbf{q}$ cyclic subspace of $QJ_T(\mathbf{q})P$; verify that all modes with energy $O(|\mathbf{q}|^2)$ are invariant under the low block; compute the compressed H , J , and T matrices and identify a common symbol $G_\eta(\mathbf{q})$; prove uniform H_{-1} tightness on the remaining current-generated sector; and check that $G_\eta(0)$ commutes with the full point group. These checks have not been completed for the full operator-valued degree-eight current sector of the explicit $\mathcal{C}_8$ lift. Consequently, the truncated equality claimed later remains conditional on this audit rather than following from row-relative form alone.

Theorem 4.4 (Isotropic transverse infrared theorem). *Assume:*

1. $G_\eta(\mathbf{q}) = \mathbf{1} + 2\eta X(\mathbf{q})$ is uniformly positive and Lipschitz near $\mathbf{q} = 0$;
2. the deformation is hydrodynamically closed in the sense of Definition 4.1;
3. its zero-momentum vector symbol is point-group scalar,

$$G_\eta(0) = \mathbf{1}_d \otimes g_\eta, \quad g_\eta \succ 0; \quad (36)$$

4. the parent density mode obeys $E_0(\mathbf{q}) \asymp \varepsilon |d(\mathbf{q})|^2$ near the origin.

Then for every normalized transverse source,

$$\left\| (d^\dagger \otimes \mathbf{1}) G_\eta(\mathbf{q}) t(\mathbf{q}) \right\| \leq C\eta |\mathbf{q}|^2, \quad (37)$$

so its hydrodynamic current H_{-1} weight satisfies

$$\langle J_\eta, H_\eta^+ J_\eta \rangle \leq C\varepsilon\eta^2 |\mathbf{q}|^2. \quad (38)$$

Equivalently, the soft spectral tail obeys

$$\sup_L \frac{1}{V_L} \langle J_{L,T}, H_L^+ \mathbf{1}_{(0,\delta]}(H_L) J_{L,T} \rangle \leq C\eta^2 \delta + o_\delta(1). \quad (39)$$

It is therefore H_{-1} -tight. The uniform Kohn kernel, the thermodynamic-first Abelian zero-frequency kernel, and the static transverse Meissner kernel exist together and satisfy

$$\boxed{D_K(t) = D_A(t) = D_M(t) = \varepsilon t(0)^\dagger (\mathbf{1}_d \otimes g_\eta) t(0) + D_{\text{reg}}(0, 0; t)}. \quad (40)$$

In particular,

$$D_M = \lambda^6 \Delta D_{\text{parent}} + O(\lambda^8 \Delta). \quad (41)$$

Proof. Quasi-locality gives $\|G_\eta(\mathbf{q}) - G_\eta(0)\| \leq C_\eta |\mathbf{q}|$. By (28) and (36),

$$(d^\dagger \otimes \mathbf{1})G_\eta(0)t = (d^\dagger \otimes g_\eta)t = 0. \quad (42)$$

Since $\|d(\mathbf{q})\| = O(|\mathbf{q}|)$, the remaining difference gives (37). Dividing the squared current amplitude by the quadratic energy proves (38). Restricting to energies at most δ gives $\varepsilon |\mathbf{q}|^2 \lesssim \delta$, hence (39); uniform tightness of the regular sector adds only $o_\delta(1)$ to the soft tail.

The static weighted residual in the closed sector is

$$\mathcal{C}_\eta(\mathbf{q}; t) = \varepsilon \left[t^\dagger G_\eta t - t^\dagger G_\eta (d \otimes \mathbf{1}) ((d^\dagger \otimes \mathbf{1}) G_\eta (d \otimes \mathbf{1}))^+ (d^\dagger \otimes \mathbf{1}) G_\eta t \right]. \quad (43)$$

The subtraction is $O(\varepsilon \eta^2 |\mathbf{q}|^2)$ and the first term converges to the right-hand side of (40). At $q = 0$ the gradient vanishes, so the same expression is the Kohn kernel. The H_{-1} -tightness criterion identifies it with the Abelian zero-frequency kernel. This proves all three limits. $\square$

Remark 4.5 (Why point-group symmetry is the natural microscopic input). For the hypercubic point group, the spatial vector representation is irreducible. Any zero-momentum row metric commuting with all signed coordinate permutations is therefore $\mathbf{1}_d$ on the spatial index, though it may be an arbitrary positive matrix on block multiplicities. Equation (36) is thus forced by a symmetric placement of identical control gadgets on every oriented bond. It is not forced in a one-dimensional construction, where there is no independent transverse channel.

Corollary 4.6 (Microscopic branch floor under an isotropic safe remainder). *For the $d \geq 2$ symmetric extension of the two-block parent, suppose the order-eight low-block remainder is a hydrodynamically closed geometric row metric satisfying (36). For a selected product-AGP composition with block densities ρ_b , the electronic-link response obeys*

$$D_K = D_A = D_M \geq 2\lambda^6 \Delta \left(1 - C\lambda^2\right) \sum_b j_b \rho_b (1 - \rho_b). \quad (44)$$

The right-hand side is positive whenever at least one selected block is nonsaturated and λ is sufficiently small. The pair-momentum superfluid weight is one quarter of (44).

5. A finite-cluster criterion for state-source safety

The abstract metric hypothesis can be tested cluster by cluster. Let D_X be the stack of all parent rows supported in a finite connected cluster X , and let

$$p_X = P_{\text{Ker } D_X}, \quad q_X = \mathbf{1} - p_X. \quad (45)$$

Proposition 5.1 (Two-sided parent-row ideal). *For a Hermitian cluster operator r_X , the following are equivalent:*

1. $p_X r_X = r_X p_X = 0$;
2. $r_X = q_X r_X q_X$;
3. *there is a Hermitian target operator Y_X such that*

$$r_X = D_X^\dagger Y_X D_X. \quad (46)$$

One explicit choice is

$$Y_X = (D_X^\dagger)^\dagger r_X D_X^\dagger, \quad \|Y_X\| \leq \|D_X^\dagger\|^2 \|r_X\|. \quad (47)$$

Proof. The equivalence of the first two statements is immediate. The third implies the first because $D_X p_X = 0$. Conversely, $D_X^\dagger D_X = q_X$, so (47) gives

$$D_X^\dagger Y_X D_X = (D_X^\dagger D_X)^\dagger r_X (D_X^\dagger D_X) = q_X r_X q_X = r_X. \quad (48)$$

The norm bound follows by submultiplicativity. $\square$

For the pair-permutation parent on a connected bounded-degree cluster, the smallest positive eigenvalue in each fixed-pair sector equals the one-particle graph gap by Aldous' spectral-gap theorem [4]. A standard graph bound then gives

$$\|D_X^\dagger\|^2 \leq C |X|^2 \quad (49)$$

(up to fixed onsite and coupling constants). Polynomial support growth is absorbed by an exponential interaction weight: for every $0 < \mu' < \mu$,

$$\sum_X e^{\mu' \text{diam } X} \|Y_X\| \leq C_{\mu-\mu'} \sum_X e^{\mu \text{diam } X} \|r_X\|. \quad (50)$$

Thus a quasi-local cluster interaction satisfying the two-sided condition assembles into a bounded quasi-local target operator. This is the precise sense in which a local row-ideal audit controls the zero-field state source uniformly in volume.

Definition 5.2 (Unsafe cluster projection). Fix the canonical connected-cluster decomposition delivered by the local Schrieffer–Wolff recursion. Define

$$\mathfrak{S}_X(r_X) = q_X r_X q_X, \quad \mathfrak{U}_X(r_X) = r_X - \mathfrak{S}_X(r_X). \quad (51)$$

The first term is state-source safe. The second contains the local ground compression and the cross terms that can be amplified by the closing parent pseudoinverse.

Passing the state-source audit does not by itself prove the isotropic theorem. One must additionally examine the action of Y_X on the current-created vector representation. A scalar convolution on row labels passes immediately; an operator-valued row metric requires a representation audit of the low-energy source sector.

Audit consequence. Part I reduces microscopic transfer to two finite questions. First, each connected Schrieffer–Wolff coefficient must have vanishing local ground compression and cross source, equivalently it must lie in the two-sided parent row ideal. Second, its induced target metric on the current-created soft sector must have point-group scalar zero-momentum symbol. Part II now applies this test to the *complete* weighted-degree-eight coefficient. The result supersedes

the earlier single-overlap warning [8]: the full unmodified coefficient is unsafe at two-pair filling, while its exact counterterm is explicit.

Part II. Complete weighted-degree-eight audit

6. Microscopic hierarchy and the degree-eight transfer question

Let $e = (x, y)$ be an oriented nearest-neighbor edge. Each endpoint carries two active blocks $a = 1, 2$ and spin σ . The crossed bridge is

$$B_e = \sum_{\sigma} \left(c_{1,y,\sigma}^{\dagger} c_{2,x,\sigma} + c_{1,x,\sigma}^{\dagger} c_{2,y,\sigma} + \text{H.c.} \right). \quad (52)$$

The microscopic control attached to e has low state g and high states m, s, a . In units of the control scale Δ , the weighted hierarchy is

$$H(\lambda) = H_0 + \lambda V_1 + \lambda^2 V_2 + \lambda^4 W_4 + \lambda^6 W_6, \quad (53)$$

where V_1 contains the same- and opposite-chirality routers, V_2 contains $bB_e(g \leftrightarrow m)$, and the screened order-four term is

$$W_4 = \frac{b^2}{\Delta_m} \sum_e B_e^2. \quad (54)$$

The degree-six term contains the base swap parent, the second matched B^2 coefficient, and the positive multiplicative interaction. The product-AGP zero manifold at total pair number n is

$$\mathcal{Z}_n = \text{span}\{|n_1, n_2\rangle : n_1 + n_2 = n\}. \quad (55)$$

The earlier microscopic theorem [7] gives

$$H_{\text{eff}}^{[7]} = \lambda^6 \Delta \left(H_{\text{base}} + \bar{\alpha} \sum_e K_e^{\dagger} K_e \right) + O_{\text{loc}}(\lambda^8 \Delta), \quad (56)$$

with complete Peierls derivative control. The exact parent has a closing spatial gap, so an $O_{\text{loc}}(\lambda^8)$ estimate alone does not transfer the response. The first task is to determine whether the canonical degree-eight coefficient preserves the zero manifold or creates a current-relevant state source.

7. Complete degree-eight resolvent enumeration

Let P fix every control in g and $Q = 1 - P$. The only low–high coupling is

$$QH(\lambda)P = \lambda^2 C_2, \quad C_2 = b \sum_e |m_e\rangle \langle G| \otimes B_e. \quad (57)$$

Inside the high block write

$$QH(\lambda)Q = H_{0,Q} + \lambda A + \lambda^2 B_Q + \lambda^4 C_Q + O(\lambda^6), \quad (58)$$

where $A = QV_1Q$, $B_Q = QV_2Q$, and $C_Q = QW_4Q$. Put $R_0 = H_{0,Q}^{-1}$. Since the outer bridges carry weight four in total, only the weighted-degree-four coefficient of the high resolvent can contribute to $R^{(8)}$.

Proposition 7.1 (Complete single-flavor class list). *The degree-four high-resolvent coefficient is*

$$R_4 = R_0 A R_0 A R_0 A R_0 A R_0 + R_0 B_Q R_0 B_Q R_0 - R_0 C_Q R_0 - \sum_{\pi \in \text{Perm}(A, A, B_Q)} R_0 \pi_1 R_0 \pi_2 R_0 \pi_3 R_0. \quad (59)$$

The mixed AAB_Q contribution vanishes between C_2 and $C_2^\dagger$. Hence

$$R^{(8)} = -C_2^\dagger R_4 C_2 = R_{A^4}^{(8)} + R_{B_Q^2}^{(8)} + R_{C_Q}^{(8)}. \quad (60)$$

No degree-six base or second-shell insertion can contribute at weighted degree eight.

Proof. The Neumann expansion of $(H_{0,Q} + X)^{-1}$ gives every ordered word whose weights sum to four, with sign $(-1)^{\text{word length}}$. These are $1 + 1 + 1 + 1$, $2 + 2$, 4 , and $1 + 1 + 2$. A state entering through C_2 has one m -control excitation. Each B_Q toggles the number of m -type control excitations by one, whereas A only routes $m \leftrightarrow s, a$. A word with one B_Q cannot return a one-excitation state to the one-excitation sector required by $C_2^\dagger$, so every AAB_Q word vanishes. A weight-six insertion would give total weight at least ten after the two outer bridges. $\square$

The zero-energy Feshbach coefficient is sufficient for the present audit. On the parent zero manifold the first possible energy shift is order λ^8 , while the energy derivative of the Feshbach map begins at order λ^4 ; energy dependence therefore changes the branch equation only at order λ^{12} .

8. Closed degree-eight source formula

Define

$$\mathcal{B}_2 = \sum_e B_e^2. \quad (61)$$

The all-filling router identities imply, on the seniority-zero bridge source,

$$C_{s,e}^2 B_e P_0 = 4Q_{s,e} B_e P_0, \quad C_{a,e}^2 B_e P_0 = 4Q_{a,e} B_e P_0. \quad (62)$$

Because $Q_{a,e} B_e P_Z = K_e P_Z = 0$, only the same-chirality channel contributes to the unsafe zero-manifold source.

Lemma 8.1 (Router source). *On the complete product-AGP manifold,*

$$R_{A^4}^{(8)} P_Z = -\frac{16b^2 s^4}{\bar{\Delta}_m^3 \bar{\Delta}_s^2} \mathcal{B}_2 P_Z. \quad (63)$$

On the complete seniority-zero space the same coefficient decomposes as

$$R_{A^4}^{(8)} = -c_s \sum_e B_e^2 + (c_s - c_a) \sum_e K_e^\dagger K_e, \quad (64)$$

where

$$c_s = \frac{16b^2 s^4}{\bar{\Delta}_m^3 \bar{\Delta}_s^2}, \quad c_a = \frac{16b^2 a^4}{\bar{\Delta}_m^3 \bar{\Delta}_a^2}. \quad (65)$$

Proof. A four-router path must alternate through m . Mixed s - a paths contain $Q_s Q_a$ on the routed source and vanish. Pure s and pure a paths give the denominators in (65). Finally $BQ_s B = B^2 - K^2$ and $BQ_a B = K^2$ on seniority zero. $\square$

The bridge-overlap term and the high-sector feedback of the matched W_4 term combine more simply than either term separately.

Proposition 8.2 (Linked commutator-square bridge metric). *For the single-flavor control,*

$$\left(R_{B_Q^2}^{(8)} + R_{C_Q}^{(8)}\right) = \frac{b^4}{\Delta_m^3} \mathcal{M}, \quad (66)$$

where

$$\mathcal{M} = \sum_e B_e^4 + \frac{1}{2} \sum_{e < f} [B_e, B_f]^\dagger [B_e, B_f]. \quad (67)$$

Only overlapping bonds contribute to the second sum.

Proof. The C_Q insertion gives

$$\frac{b^4}{\Delta_m^3} \sum_{e,f} B_e B_f^2 B_e. \quad (68)$$

The two-control bridge path gives, for each unordered pair,

$$-\frac{b^4}{2\Delta_m^3} \{B_e, B_f\}^2. \quad (69)$$

For Hermitian E, F ,

$$EF^2 E + FE^2 F - \frac{1}{2} \{E, F\}^2 = \frac{1}{2} [E, F]^\dagger [E, F]. \quad (70)$$

The diagonal terms in (68) are B_e^4 . If the active supports are disjoint, the commutator vanishes, so the result is a linked local interaction rather than a disconnected product. $\square$

Combining the two results gives the complete source coefficient.

Theorem 8.3 (Exact degree-eight source formula). *For the unmodified single-flavor screened microscopic family,*

$$R^{(8)} P_Z = \frac{b^2}{\Delta_m^3} \left[b^2 \mathcal{M} - \frac{16s^4}{\Delta_s^2} \mathcal{B}_2 \right] P_Z. \quad (71)$$

The closed-shell Kramers completion replaces $\mathcal{M}$ by

$$\mathcal{M}_\Theta = \frac{1}{2} \sum_e B_e^4 + \frac{1}{2} \sum_{e < f} [B_e, B_f]^\dagger [B_e, B_f]. \quad (72)$$

The inter-bond coefficient is unchanged; double excitation of the two partners halves only the local B_e^4 coefficient.

Proof. The single-flavor result is Lemma 8.1 and Proposition 8.2. In the closed shell, each of two independent Kramers channels has bridge amplitude $b/\sqrt{2}$. Summing the two same-channel local paths gives $2(b/\sqrt{2})^4 = b^4/2$. For two distinct bonds there are four partner pairs, each with weight $b^4/4$; their total leaves the single-flavor commutator-square coefficient unchanged. $\square$

9. Two-pair all-filling obstruction

A one-pair calculation alone is misleading. On a three-cell ring, the one-pair compression of (71) is proportional to the all-ones matrix and can be canceled accidentally by

$$b_* = \frac{2s^2}{\bar{\Delta}_s}. \quad (73)$$

The cancellation does not extend to two pairs.

Let

$$|0, 2\rangle = \frac{1}{\sqrt{3}} \sum_{0 \leq x < y \leq 2} d_{2,x}^\dagger d_{2,y}^\dagger |0\rangle, \quad |\Phi\rangle = d_{1,0}^\dagger d_{1,1}^\dagger |0\rangle. \quad (74)$$

Theorem 9.1 (Two-pair degree-eight obstruction). *On the three-cell two-pair sector,*

$$\langle \Phi | \mathcal{B}_2 | 0, 2 \rangle = 0, \quad \langle \Phi | \mathcal{M} | 0, 2 \rangle = 8\sqrt{3}, \quad \langle \Phi | \mathcal{M}_\Theta | 0, 2 \rangle = 4\sqrt{3}. \quad (75)$$

Consequently, for $b \neq 0$, neither the single-flavor nor the closed-shell Kramers unmodified family preserves the complete product-AGP zero manifold at weighted degree eight. No scalar retuning of b , s , a , or an order-eight adjustment of the matched B^2 coefficient can cancel the full all-filling source while the required degree-four cancellation is retained.

Proof. Every term in $\mathcal{B}_2$ contains two one-electron bridge hops and can transfer at most one onsite pair between blocks. It therefore has zero matrix element from $|0, 2\rangle$ to $|\Phi\rangle$, which has both pairs transferred. The commutator $[B_e, B_f]$ of two number-conserving fermion bilinears is itself a bilinear. Hence $[B_e, B_f]^\dagger [B_e, B_f]$ contains at most two one-electron transfers after normal ordering and likewise cannot connect the two witness states. Thus

$$\langle \Phi | \frac{1}{2} \sum_{e < f} [B_e, B_f]^\dagger [B_e, B_f] | 0, 2 \rangle = 0. \quad (76)$$

Only the local B_e^4 terms contain the four transfers required to move both onsite pairs. Direct fermionic action on the three translated bonds gives $8\sqrt{3}$. Equation (76) then makes the Kramers value exactly one half, because (72) halves only the local B_e^4 coefficient. The router term in (71) is proportional to $\mathcal{B}_2$ and cannot affect this matrix element. The separate vanishing in (76) is also checked by the retained structure certificate. $\square$

Remark 9.2 (Analytic scope of the no-go). Below the remote control gap, the finite-volume Feshbach map is analytic in λ . Preservation of the complete product-AGP manifold for every sufficiently small coupling would therefore force each weighted coefficient to preserve it. The nonzero matrix element in (75) rules out such an analytic all-composition transfer. It does not exclude a nongeneric isolated finite value of λ at which higher-order leakage might accidentally cancel the degree-eight term; no such isolated cancellation is asserted or observed here.

The obstruction is stronger than a poor parameter choice. Writing

$$R^{(8)} P_Z = b^2 (U + b^2 V) P_Z, \quad (77)$$

U and V are exactly antiparallel in the one-pair sector, but not at higher filling. The accompanying exact finite-Fock-space certificate finds the following minimum over every scalar b for the representative parameters $\bar{\Delta}_m = 10$, $\bar{\Delta}_s = \bar{\Delta}_a = 8$, $s = 0.9$, $a = 0.45$.

Pairs	$\cos_F(U, V)$	best b	minimum source norm of $U + b^2V$
1	-1	0.2025	$< 3 \times 10^{-12}$
2	-0.9058554866	0.1181008620	8.6587×10^{-4}
3	-0.9486832981	0.1181008615	8.6587×10^{-4}

The numerical values certify the finite matrices; Theorem 9.1 is the analytic reason the higher-filling residual cannot vanish.

10. Explicit degree-eight counterterm and truncated transfer

The negative result identifies a simple finite-order repair. Define the active counterinteraction

$$\mathcal{C}_8 = \frac{16b^2s^4}{\bar{\Delta}_m^3 \bar{\Delta}_s^2} \sum_e B_e^2 - \frac{b^4}{\bar{\Delta}_m^3} \mathcal{M} \quad (78)$$

for the single-flavor model, and replace $\mathcal{M}$ by $\mathcal{M}_\Theta$ for the closed shell.

Corollary 10.1 (Explicit response-safe degree-eight completion). *Adding $\lambda^8 \Delta \mathcal{C}_8$ cancels the complete degree-eight source on every product-AGP filling. On seniority zero, the surviving degree-eight term is*

$$(c_s - c_a) \sum_e K_e^\dagger K_e, \quad (79)$$

which is row-relative and positive whenever $s^2/\bar{\Delta}_s > a^2/\bar{\Delta}_a$.

Proof. Use (64), (66), and (78). The B^2 and $\mathcal{M}$ pieces cancel identically, leaving (79). $\square$

This is the complete, model-specific replacement for the earlier abstract “remove the unsafe projection and average the safe metric” construction. The counterterm cancels the *entire* degree-eight source formula, not only the isolated two-control overlap. It is finite range because $[B_e, B_f] = 0$ on disjoint supports, and its surviving $K_e^\dagger K_e$ contribution is manifestly in the two-sided parent-row ideal of Proposition 5.1.

Proposition 10.2 (Locality and Peierls covariance of the explicit counterterm). *The counterinteraction (78) is finite range. Replacing each bridge by its fully covariant link-field version $B_e(A)$ gives a Peierls-covariant interaction $\mathcal{C}_8(A)$ whose first two field derivatives have volume-uniform local interaction norms. Inserted in the microscopic Hamiltonian with coefficient $\lambda^8 \Delta$, it changes the canonical low-block coefficient at weighted degree eight by exactly $\mathcal{C}_8$; conjugation corrections from a fixed degree-seven Schrieffer–Wolff unitary begin at higher weighted degree.*

Proof. The commutator term is supported only on overlapping bonds because $[B_e, B_f] = 0$ for disjoint active supports. The remaining terms are single-bond polynomials. Covariant link substitution preserves these support bounds, and two field derivatives only insert bounded displacement factors. Finally, an order-eight active term is acted on by the lower-order block-diagonalizing unitary as the identity plus positive powers of λ , so its leading low-block contribution is its own order-eight coefficient. $\square$

Corollary 10.3 (Conditional response transfer through weighted degree eight). *The completed coefficient satisfies the exact two-sided row-ideal criterion on the complete product-AGP manifold. Suppose, in addition, that its point-group-complete $d \geq 2$ Peierls lift is hydrodynamically closed*

on the current-created soft sector in the sense of Definition 4.1. Point-group symmetry then makes its zero-momentum spatial vector symbol scalar, and

$$D_K^{[8]} = D_A^{[8]} = D_M^{[8]} = \lambda^6 \Delta D_{\text{parent}} + O(\lambda^8 \Delta). \quad (80)$$

Under $s^2/\bar{\Delta}_s > a^2/\bar{\Delta}_a$, the explicit degree-eight multiplicative-row shift is positive.

Proof. The cancellation and surviving row-relative term are Corollary 10.1. The additional closure hypothesis identifies that row metric with the soft current representation used in Theorem 4.4; point-group completeness then enforces its scalar zero-momentum spatial factor. The isotropic theorem proves (80). $\square$

The completion is not two-electron-only: after normal ordering, B_e^4 contains higher-body active interactions. Synthesizing (78) from the restricted exchange/correlated-hopping tensor of the original microscopic model, or recursively completing every higher weighted order with a convergent local map, remains open.

Part III. Exact dark branch and finite-coupling Meissner floor

11. The exact coherent dark branch

The failure of the complete product manifold does not imply that every microscopic paired branch is lost. Define

$$\eta_-^+ = \eta_1^+ - \eta_2^+, \quad |\Omega_n^-\rangle = \mathcal{N}_{L,n}(\eta_-^+)^n |0\rangle. \quad (81)$$

In the normalized product-composition basis,

$$|\Omega_n^-\rangle = \frac{1}{\sqrt{\binom{2L}{n}}} \sum_{n_1+n_2=n} (-1)^{n_2} \sqrt{\binom{L}{n_1} \binom{L}{n_2}} |n_1, n_2\rangle. \quad (82)$$

Proposition 11.1 (Exact bridge darkness). *For every edge e and every filling,*

$$[B_e, \eta_-^+] = 0, \quad B_e |\Omega_n^-\rangle = 0. \quad (83)$$

Within $\mathcal{Z}_n$, the common kernel of all B_e is one dimensional and is spanned by $|\Omega_n^-\rangle$ on a connected lattice.

Proof. Suppress spin and write the one-body bridge matrix as $h_e = \tau_x \otimes A_e$, where $A_e = A_e^T$ is the endpoint adjacency and τ_x acts on the two blocks. The singlet pairing matrix of η_-^+ is $F = \tau_z \otimes \mathbf{1} \otimes i\sigma_y$. For a one-body operator $c^\dagger h c$,

$$[c^\dagger h c, \eta_F^+] = \frac{1}{2} c^\dagger (h F + F h^T) c^{\dagger T}. \quad (84)$$

Since $\{\tau_x, \tau_z\} = 0$, the commutator vanishes. Also $B_e |0\rangle = 0$, proving the second identity.

For uniqueness, put $r = n_1$ and $n_2 = n - r$. Direct hard-core reduction and AGP combinatorics give, on the periodic L -edge lattice,

$$\langle r, n-r | \sum_e B_e^2 | r, n-r \rangle = 4n - \frac{8r(n-r)}{L}, \quad (85)$$

$$\langle r+1, n-r-1 | \sum_e B_e^2 | r, n-r \rangle = \frac{4}{L} \sqrt{(r+1)(L-r)(n-r)(L-n+r+1)}. \quad (86)$$

The product-AGP composition states are invariant under independent permutations of the L sites within each block. Consequently, the compression of B_e^2 is identical for every edge e . For an arbitrary graph with $|E| > 0$, the compression of $\sum_{e \in E} B_e^2$ is therefore $|E|/L$ times (85)–(86). Its off-diagonal entries are strictly positive at every interior point of the allowed composition interval, so the compression is an irreducible Hermitian tridiagonal matrix. Such a Jacobi matrix has simple eigenvalues. Equation (83) supplies a zero vector, hence the common bridge kernel in $\mathcal{Z}_n$ is one dimensional. In particular, no translation-invariance assumption is needed for the connected-lattice statement. $\square$

Corollary 11.2 (All-orders zero-field eigenbranch). *In the screened microscopic family, with all controls in their g states,*

$$|\Omega_n^-\rangle \otimes |G\rangle \quad (87)$$

is an exact zero-energy eigenstate at $A = 0$ for every finite coupling λ for which the Hamiltonian is defined.

Proof. The base parent annihilates every vector in $\mathcal{Z}_n$. Every bridge vertex contains B_e and annihilates (87); every direct shell term contains B_e^2 and also annihilates it. Router vertices act only after a control has left g , which never occurs on the dark state. The screened monopole terms cancel by construction. $\square$

The corollary does not by itself prove that the exact eigenbranch is the ground state of the original matched model. The finite-order matching does not make the complete microscopic Hamiltonian positive semidefinite. The next construction supplies that missing property without using a closing-gap perturbation theorem.

12. Positive-semidefinite finite-coupling completion

Consider one control edge. After separating the low control state g from the complete high control space, write its local Hamiltonian as

$$h_e(0) = \begin{pmatrix} \beta B_e^2 & t_B B_e P_m^\dagger \\ t_B P_m B_e & D_e \end{pmatrix}, \quad D_e \succeq d_* \mathbf{1} > 0, \quad (88)$$

where P_m inserts the bridge source into the m component of the high block. The router is contained in D_e .

Theorem 12.1 (Strict overcompensation). *If*

$$\beta > \frac{t_B^2}{d_*}, \quad (89)$$

then $h_e(0) \succeq 0$. Moreover, every zero vector of $h_e(0)$ has its control in g and satisfies $B_e \psi = 0$.

Proof. The Schur complement of D_e is

$$S_e = \beta B_e^2 - t_B^2 B_e P_m^\dagger D_e^{-1} P_m B_e. \quad (90)$$

Since $D_e^{-1} \preceq d_*^{-1} \mathbf{1}$,

$$S_e \succeq \left(\beta - \frac{t_B^2}{d_*} \right) B_e^2 \succeq 0. \quad (91)$$

The standard block Schur criterion gives $h_e \succeq 0$. Equality in the strict bound forces $B_e \psi = 0$, and the high component $-D_e^{-1} t_B P_m B_e \psi$ then vanishes. $\square$

Define the exact finite-coupling family

$$H_{\text{PSD}}(0) = \lambda^6 \Delta H_{\text{base}} + \sum_e h_e(0), \quad (92)$$

with the screened density channel and the strict coefficient (89).

Corollary 12.2 (Exact fixed-number ground state). *On a connected lattice, $H_{\text{PSD}}(0)$ is frustration free and has the unique fixed-2n zero state (87) at every interior filling.*

Proof. Every term is positive and annihilates the dark state. Conversely, a global zero vector lies in the base-parent zero manifold and, by Theorem 12.1, obeys $B_e \psi = 0$ on every edge with every control in g . Proposition 11.1 gives uniqueness. $\square$

Under the complete Peierls lift, each single-edge block $h_e(A_e)$ is unitarily equivalent to its zero-field version by a gauge conjugation of the active endpoint modes and the control source. The conjugating unitary need not be supported on the edge alone, and no claim is made that a nonzero loop flux is globally pure gauge; only the termwise unitary equivalence of $h_e(A_e)$ is used. Hence $h_e(A_e) \succeq 0$ for every edge phase, and the full family remains a sum of local positive terms. This is a nonperturbative microscopic parent rather than a Schrieffer–Wolff approximation.

13. Finite-coupling microscopic Meissner floor

Let the spatial lattice have $d \geq 2$, include every point-group-related edge orientation, and couple a physical transverse edge field. Factor the positive Hamiltonian into a direct-sum row map

$$\mathcal{D}_{\text{full}} = \begin{pmatrix} \lambda^3 \sqrt{\Delta} \mathcal{D}_{\text{sw}} \\ \mathcal{D}_{\text{gad}} \end{pmatrix}, \quad \mathcal{S}_{\text{full}} = \begin{pmatrix} \lambda^3 \sqrt{\Delta} \mathcal{S}_{\text{sw}} \\ \mathcal{S}_{\text{gad}} \end{pmatrix}. \quad (93)$$

For every divergence-free field, the graph-Hodge identity of [6] gives

$$\mathcal{D}_{\text{sw}}^\dagger \mathcal{S}_{\text{sw}} = 0. \quad (94)$$

Theorem 13.1 (Finite-coupling transverse response floor). *For the exact PSD microscopic family and the selected dark branch, every finite-volume static curvature and imaginary-frequency transverse kernel satisfies*

$$\mathcal{K}_{L,T}(q, i\zeta) \geq \mathcal{C}_{L,T}(q) \geq \lambda^6 \Delta \|\mathcal{S}_{\text{sw},T}(q)\|^2, \quad \zeta \geq 0. \quad (95)$$

If both block swap couplings are j_1, j_2 , then per spatial volume

$$\boxed{D_{\text{sw},L}^{(-)} = (j_1 + j_2) \frac{n(2L - n)}{L(2L - 1)}}. \quad (96)$$

Consequently, with $n/(2L) \rightarrow \rho \in (0, 1)$,

$$\boxed{\begin{aligned} \liminf D_K &\geq D_*, & \liminf D_A &\geq D_*, & \liminf D_M &\geq D_*, \\ D_* &= 2\lambda^6 \Delta(j_1 + j_2) \rho(1 - \rho) > 0 \end{aligned}} \quad (97)$$

The pair-phase superfluid-weight floor is one quarter of (97).

Proof. The static positive-square curvature is the least-squares residual

$$\mathcal{C} = \min_X \|\mathcal{S}_{\text{full}} - \mathcal{D}_{\text{full}} X\|^2. \quad (98)$$

By (94),

$$\|\mathcal{S}_{\text{sw}} - \mathcal{D}_{\text{sw}} X\|^2 = \|\mathcal{S}_{\text{sw}}\|^2 + \|\mathcal{D}_{\text{sw}} X\|^2, \quad (99)$$

so no state correction, including one using the gadget sector, can cancel the transverse swap source. The finite-frequency kernel exceeds the static residual by the positive spectral defect, proving (95).

The composition probabilities in (82) form a hypergeometric distribution. Averaging the exact blockwise swap response $2j_a n_a (L - n_a) / [L(L - 1)]$ gives

$$\mathbb{E}[n_a(L - n_a)] = \frac{n(L - 1)(2L - n)}{2(2L - 1)}, \quad (100)$$

which yields (96). Taking any admissible thermodynamic, zero-frequency, and transverse zero-momentum limits gives (97). $\square$

Remark 13.2 (What the floor does and does not identify). The theorem is already a genuine microscopic zero-frequency Meissner statement: the static transverse response is uniformly positive at every allowed q , and the thermodynamic-first imaginary-frequency kernel has the same positive floor. It does not assert that the additional gadget contribution has identical Kohn, Abelian, and transverse limits. Proving equality for the *total* response still requires the H_{-1} tightness and point-group closure conditions developed in Part I.

14. Conclusion and remaining microscopic problem

The merged analysis separates three statements that had previously been conflated.

First, the infrared transfer problem has a clean functional-analytic answer. A local remainder must control both the zero-field state source and the current source. The two-sided row ideal eliminates the former. Target-metric factorization controls the latter uniformly through a closing parent gap, but only point-group isotropy converts bounded soft-mode weight into the H_{-1} tightness required for exact Kohn–Abelian–Meissner identification. The anisotropic counterexample shows that this distinction is sharp.

Second, the unmodified matched microscopic family fails the stronger all-composition objective. The complete degree-eight source contains a linked fourth-power bridge metric that cannot be canceled by any scalar B^2 retuning at two-pair filling. The no-go survives the closed-shell Kramers completion. The explicit counterterm (78) repairs the entire coefficient through weighted degree eight and leaves a positive parent-row correction, thereby providing a precise finite-order state-source completion rather than an abstract existence claim. Equality of the three truncated

response limits then follows under the separate hydrodynamic-closure and point-group symbol audit of Part I.

Third, a different finite-coupling objective is already achieved. The coherent dark AGP is annihilated by every primitive bridge. Strict local Schur-complement overcompensation makes it the exact ground branch of a positive-semidefinite microscopic family, and the swap backbone supplies the nonzero transverse floor (97). This is a genuine zero-temperature, zero-frequency microscopic Meissner statement for a selected branch, although it is a lower-bound theorem rather than a complete equality formula for the gadget response.

The remaining all-composition problem is now an analytic synthesis problem, not a thermodynamic numerical scan: realize $\mathcal{C}_8$ using restricted microscopic two-electron ancillary channels and determine whether the resulting sequence of row-ideal counterterms admits a convergent local Feshbach or Schrieffer–Wolff normal form. Independently, applying the Part I isotropy theorem to every extra row of the positive dark-branch family would upgrade its strict floor to a full equality theorem for the total response.

A. State-space derivation of the target-metric formulas

Let $\tilde{H} = (\varepsilon/2)\tilde{D}^\dagger\tilde{D}$ with $\tilde{D} = G^{1/2}D$. Acting on the zero manifold,

$$JP = \frac{\varepsilon}{2}\tilde{D}^\dagger\tilde{S}, \quad TP = \varepsilon\tilde{S}^\dagger\tilde{S}, \quad \tilde{S} = G^{1/2}S. \quad (101)$$

The imaginary-frequency Kubo kernel is

$$\mathcal{K}(i\zeta) = \varepsilon\tilde{S}^\dagger\tilde{S} - 2PJQ\tilde{H}(\tilde{H}^2 + \zeta^2)^{-1}QJP. \quad (102)$$

Using the singular-value decomposition of $\tilde{D}$ reduces (102) to (13). At $\zeta = 0$, the positive singular modes cancel and only $\text{Ker } \tilde{D}^\dagger$ remains. The defect bound follows from

$$0 \leq \frac{\zeta^2}{E(E^2 + \zeta^2)} \leq \frac{1}{E}. \quad (103)$$

B. Exact one-mode anisotropic calculation

For (23)–(24), put $u = e^{iq} - 1$. Then

$$d^\dagger Gd = |u|^2, \quad d^\dagger Gt = 2\eta u^*, \quad t^\dagger Gt = 1. \quad (104)$$

The static weighted residual is

$$1 - \frac{|2\eta u|^2}{|u|^2} = 1 - 4\eta^2. \quad (105)$$

The state energy and current are $E = |u|^2/2$ and $J = \eta u^*$, which give (26) and (27). No asymptotic approximation is used.

C. Reproducibility and certificates

The consolidated archive contains eight deterministic verification programs.

1. `verify_hminusone_transfer.py` checks the target-space and state-space response formulas, the metric sandwich, the gap-independent defect bound, the local row-ideal factorization, and the anisotropic example.
2. `verify_two_control_overlap.py` independently extracts the neighboring-control fourth-order Feshbach coefficient and its one-pair compression.
3. `audit_order8_all_fillings.py` evaluates every retained weighted-degree-eight source class on every product-AGP filling of the three-cell ring.
4. `verify_order8_structure.py` checks the router source, linked commutator-square identity, mixed-class vanishing, the separate zero commutator-square witness matrix element, the remaining two-pair amplitudes, and the exact counterterm.
5. `analyze_order8_tuning.py` certifies the accidental one-pair scalar cancellation and the nonzero higher-filling residual.
6. `verify_kramers_feshbach.py` independently extracts the single-flavor and closed-shell coefficients from finite-control Feshbach maps with noncommuting active matrices.
7. `analyze_kramers_order8.py` checks the Kramers modification of the local and inter-bond bridge metric.
8. `verify_dark_agp_branch.py` checks exact bridge darkness, the one-dimensional composition kernel, the Jacobi entries, and the finite-volume swap-floor formula.

All retained outputs end in **PASS**, **OVERALL: PASS**, or **AUDIT IDENTITIES: PASS**. These calculations certify the finite matrices and algebraic identities used in the paper; they do not replace the analytic arguments.

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